ROBLOX STUDIO EXPERT SKILLS · TEAM SKILL 5 OF 14

Player Data

Backend engineer, live-ops programmer

`scenario-roblox-data`

Builds a player data layer that survives crashes, failed writes and server hops: session-locked profiles, safe migrations, one grant per purchase, proven against a simulated multi-server backend.



Thumbnail of Rileybytes' ProfileStore Developer Products tutorial; the skill takes its saved-before-granted rule (00:29:08).

USE IT WHEN

- Players report data not saving, lost progress, rollbacks or duplicated items
- A Developer Product grants twice, or a trade loses or copies items
- Setting up ProfileStore or migrations, or testing data code in an unpublished place

WHAT THE EXPERTS INSIST ON

- One ProfileStore profile per player holds the session lock; a profile load over 5 s is a bug. Ioleris (MAD STUDIO, ProfileStore author)
- UpdateAsync beats SetAsync only when the transform reads the old value; session locks are what stop duplication. Suphi Kaner (author, Suphi's DataStore Module)
- Return PurchaseGranted only once the saved profile holds that PurchaseId; every other path returns NotProcessedYet. Rileybytes (Roblox educator)
- A failed write may have landed: read with UseCache false before redoing it, and retry each key in order. Roblox (data store best practices)
- Publish every place of a multi-place game together: an old server reading newer data errors. CJ_Oyer (Roblox Accelerator alumnus)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Probe the environment.** **Gate:** Service error texts recorded; real store or mock chosen from them.
2. **Design schema and budget.** **Gate:** Profile under 25% of 4,194,304 characters; budget ok at peak CCU.
3. **Wire session-locked profiles.** **Gate:** Sessions suite passes; a playtest shows the profile loaded with DataStoreState NoAccess.
4. **Version the schema.** **Gate:** Old data migrates once, before Reconcile; newer data is refused untouched.
5. **Grant purchases once.** **Gate:** Receipts suite passes through failed writes, crashes and server changes.
6. **Trades, stores and teleports.** **Gate:** Trade suite passes at every crash stage; keyqueue and teleport suites pass.
7. **Audit, prove and report.** **Gate:** Zero audit errors; suites pass offline and in Studio; published-only gaps listed.

KEY NUMBERS

Server read or write budget	**60 + 40 x players per minute**
Experience budget per minute	**read 300 + 40 x CCU, write 300 + 20 x CCU**
Value size per key	**4,194,304 characters; key, name and scope 50**
ProfileStore autosave, lock steal, assume dead	**300 s, 40 s, 630 s**
Naive retries, 30% of writes landing	**39 stored for 30 operations (mock run)**
Versions kept per key	**one per UTC hour, for 30 days after overwrite**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- run_offline.sh: all Lune suites pass; rbx_check zero errors in kit, place, snippets.
- run_live.py: the same suites pass in Studio Edit, plus the playtest checks.
- DataAudit: zero error findings outside test fixtures.
- Console: one 'Roblox API services unavailable' line in an unpublished playtest, no save errors.

HANDOFFS

Receives Data service slot from architecture, product ids from monetization, validated trade and shop requests from networking.

Delivers Values as Attributes or signals to UI and gameplay, the Receipts handler table, Lune suites for CI.

TOOLKIT `RBX_DATA.PY`

- `budget()`: DataStore request budget check at peak CCU
- `probe_services()`: which data services answer in Edit and playtest
- `audit_place()`: offline DataAudit of a place file

MAIN EXPERT SOURCES

Ioleris (MAD STUDIO)
ProfileStore author, code pinned to one commit, ProfileStore documentation and source

Suphi Kaner
author of Suphi's DataStore Module (partisan comparison), DataStore2 vs ProfileService vs Suphi's DataStore Module

Rileybytes
Roblox educator (unverified), Sell Developer Products EASILY with ProfileStore

Roblox Creator Docs
official documentation, Data store best practices, error codes and limits

STATUS

Tests 70 Lune tests pass

In app 26 live tests pass in Roblox Studio 0.743

G3 data

without skills	9/15
with skills	15/15